

SHREE VARAA MANGAI V

+1(904)420-9650 • shreevaraamangai@gmail.com • [LinkedIn](#) • [GitHub](#) • [Portfolio](#) • [Medium](#)

SKILLS

ML / LLM & AI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, LangChain, LlamaIndex, OpenAI/ Gemini API, Supabase, Pinecone, Gradio, Streamlit, CUDA, AWS Bedrock
Full-Stack Engineering: Python, TypeScript, JavaScript, Java, C/C++, Dart · React.js, Node.js, Flask, FastAPI, PostgreSQL, MongoDB, Firebase, SQL, REST APIs, Swagger
Deployment & Tools: Docker, Git, Linux, CI/CD Pipelines, Azure, Postman, Weights & Biases(wandb), SageMaker
Data & Visualisation: PowerBI, Tableau, Matlab, Chart.js, Microsoft Office, Google Workspace, Canva

WORK EXPERIENCE

ML Engineering Co-op | University of Florida- IPPD | AGIS Inc.

Florida, US | Aug 2025 - Apr 2026

It's an enterprise AI company that builds custom AI-powered tools for business automation.

- Ran the 2D Gaussian Splatting (2DGS) reconstruction stage end-to-end on **NVIDIA B200 GPU** clusters, using SAM2-based zero-shot object segmentation, COLMAP feature extraction/pose estimation, and Gaussian optimization, achieving **32.72 dB PSNR** / 0.0059 L1 render error.
- Benchmarked end-to-end output against Kiri Engine on identical inputs: delivered a Unity-ready rigged asset in **8 min vs. Kiri's 15 min** for a raw PLY export, a **47% latency reduction** with richer output.
- Profiled GPU workloads on NVIDIA B200 clusters using PyTorch/CUDA tooling, optimizing inference or reconstruction latency through batching, memory-aware preprocessing, and throughput benchmarking.

Machine Learning Researcher | University of Florida- Trustworthy Engineered Autonomy Lab

Florida, US | Sep 2025 - Apr 2026

Research lab focused on reliable and interpretable AI systems. This role centred on benchmarking hallucination in text-to-video generative models.

- Designed masked prompt reconstruction experiments for hallucination detection in text-to-video models, generating 15,000+ videos using text-to-video models Wan 2.1 and HunyuanVideo on B200 GPUs across T2VCompBench and ViBe prompts. Built a token-level detector that conditions a frozen **VLM (Qwen3-VL-30B)** on video keyframes and, via teacher-forced token rank rather than direct classification, measures how well each masked prompt token can be recovered from the visual evidence.
- Benchmarked the method against **12+ baselines** including direct VLM classification, self-consistency, and teacher-forced likelihood scoring, achieving the strongest prompt inconsistency detection across all six metrics on T2VCompBench with Wan 2.1 (**Balanced Accuracy .645, AUPRC .739, AUROC .687**).

Global People Analytics Intern | Ford Motor Company

Tamil Nadu, IN | Aug 2023 - Oct 2023

Ford's Global People Analytics team uses data-driven methods to inform HR strategy. This internship focused on workforce forecasting and budget planning.

- Conducted statistical analysis and forecasting on time series forecasting models such as Autoregressive Integrated Moving Average (ARIMA), Vector Autoregression (VAR), and Vector Error Correction Model (VECM) and non-time series models, such as Lasso and Ridge regression, to determine optimal female incumbency.
- Developed ETL pipelines in Python to predict salary costs using workforce datasets containing 1,000+ employee records, improving **model accuracy by 20%** through exploratory data analysis, feature engineering, and data quality improvements.

Software Engineer Intern | Spacescan Ltd.

Ontario, Canada | Sep 2022 - Jan 2023

Spacescan is a blockchain explorer and analytics platform. This internship involved full-stack development of the consumer-facing web and mobile application.

- Built and maintained responsive, component-based React.js interfaces integrating Swagger-documented REST APIs, reducing code duplication through reusable UI components and robust state management.
- Extended the React Native mobile application with new features, collaborating on frontend architecture and software design decisions to maintain consistent UI and interaction patterns across web and mobile.

PROJECTS

CareMind: Agentic Clinical RAG Assistant | [GitHub-Link](#)

May - Ongoing 2026

- Building an agentic clinical RAG system with FastAPI, LangGraph, Supabase vector search, Redis, NVIDIA NIM(nemotron-3-ultra-550b-a55b) and MCP tools to route queries across document QA, report comparison, and medical education workflows with cited evidence synthesis.
- Developing a RAG evaluation and observability pipeline using PubMedQA-style biomedical QA, Prometheus, and Grafana to measure Recall@K, Precision@K, MRR, faithfulness, latency, and TTFT; improved synthetic clinical QA generation recall from **0.598 to 0.764** and faithfulness support rate from **0.505 to 0.677**.

Tech Stack: Python, FastAPI, LangGraph, Supabase/pgvector, Redis, NVIDIA NIM, MCP, Prometheus, Grafana, Next.js, TypeScript

Multi-Image Super Resolution for Prostate MRI (CNN, SRGAN, Diffusion Models) | [GitHub-Link](#)

Nov - Dec 2025

- Trained and benchmarked **three architectures**: a 5-block Residual Interpolation CNN, SRGAN (PatchGAN discriminator with adversarial/L1 loss), and a **Fast-DDPM diffusion model** (10-step DDIM, non-uniform β schedule) on 40,532 slice-pairs to reconstruct missing prostate MRI through-plane slices.
- SRGAN led on numerical accuracy (29.045 dB PSNR, 0.850 SSIM vs. CNN's 29.027 dB/0.843), while Fast-DDPM traded **PSNR (27.019 dB, 0.801 SSIM)** for the strongest anatomical continuity, **at 24x the inference latency** (10.5ms vs. 0.4-0.6ms).

Tech Stack: PyTorch, CNN, GAN, Diffusion Models, OpenCV, NumPy

CognitoMap: Automated Question Classification using Bloom's Taxonomy | [GitHub-Link](#)

Feb - Apr 2024

- Built a scalable ETL pipeline extracting and transforming questions from PDFs into MongoDB, achieving **94% accuracy** in question extraction using the Gemini 1.0 Pro model and classified questions into Bloom's Taxonomy levels using Sentence Transformer embeddings with Pinecone vector retrieval.
- Delivered a full-stack dashboard (Flask, React.js, Chart.js) giving educators real-time insights into the cognitive distribution of their assessments across Bloom's Taxonomy levels.

Tech Stack: Flask, React, MongoDB, Pinecone, Chart.js, TypeScript

CERTIFICATIONS

- **Building with the Claude API:** Anthropic (2026) · **Build & Orchestrate Agents with MS Foundry:** Microsoft (2026) · **Fundamentals of Deep Learning:** NVIDIA (2024) · **Fine-Tuning LLMs for Cybersecurity:** LinkedIn Learning (2025) · **Tesla Stock Price Prediction / Google Cloud Fundamentals:** Coursera (Dec 2023 / Aug 2022)

EDUCATION

Master of Science in Artificial Intelligence Systems | GPA: 3.90 / 4.0

Aug 2024 - May 2026

University of Florida, Gainesville, FL

Relevant Coursework: Deep Learning in Medical Image Analysis, ML for AI Systems, Applied Deep Learning, AI Ethics, Computational Photography, Computer Vision, NLP

Bachelor of Technology in Information Technology | GPA: 8.43 / 10

Sep 2020 - May 2024

Loyola-ICAM College of Engineering and Technology, India (affiliated under Anna University)

Relevant Coursework: OOP Principles, Computational Intelligence, Computer Architecture, Cryptography & Network Security, Design and Analysis of Algorithms, Big Data Analytics, Linear Algebra, Statistics, Database Management Systems, Data Structures